Comp-4400: Assignment 2: Scheme (3 points)

Last Name	First Name	ID	Email ID	Signature	Marks	Marked by
				Lucas Zenner		

Copy/paste your code here

Assignment specification

You will define a calculator for simple arithmetic expressions in Scheme. For example, when typing

```
(calculator '(1 + 2))
```

Scheme interpreter will print 3. The calculator will accept expressions of any length, and the expression associates to the left. For example, in

```
(calculator '(1 + 1 - 2 + 3))
```

the expression is interpreted as `(((1 + 1) - 2) + 3))`, hence the result is 3. If you interpret it as right associative `(1 + (1 - (2 + 3)))` and resulting in -3, you will have one point deducted.

The syntax for the expression is:

```
exp ::= number | exp op exp
op  ::= + | -
```

For example, 12, $2+3$, and $2+3-4$ are valid expressions. To make the assignment easier, we allow only plus and minus operators.

Marking

If you evaluate the expression using left associativity, you will be awarded one bonus mark. For example, the above example should be interpreted as `((1 + 1) - 2) + 3`, therefore the result is 3. In your submission, please indicate whether it is left associative or right associative. You only need to submit one version of your program, i.e., either left associative or right associative.

You can assume that the input is a correct arithmetic expression. We will not test invalid cases like:

```
(calculator '(1 + ))
```

or

```
(calculator '(- 1 + 2))
```